

# Quality Assurance Engineering Intern - Assessment

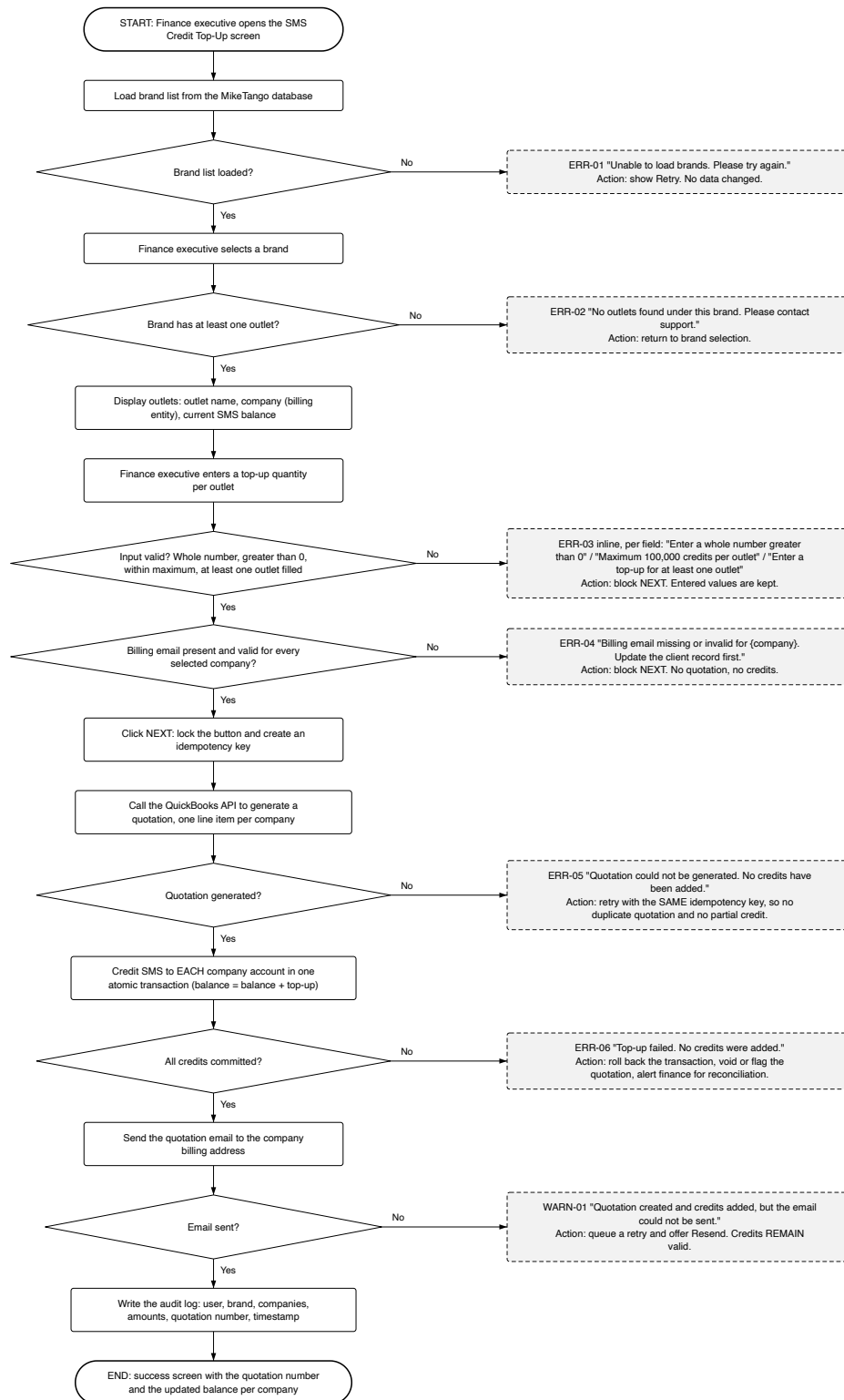
SMS Credit Reload feature (MikeTango / GoSMS) | Syed Muhammad Aziz Ghaus | azizghaus14@gmail.com | +60 17-414 7089

Assessment 1 describes the SMS Credit Reload feature and how it is intended to work. This document is the response to Assessment 2: Part 1 is the flowchart for the feature including error handling, and Part 2 is the set of test cases, supported by a state transition model that shows the coverage is complete.

## 1. Assumptions

- SMS credits belong to a company (billing entity), not to a brand. One brand can span several companies, so a single submission may bill more than one entity.
- The quotation is the commit point: credits are added only once QuickBooks confirms the quotation was generated.
- Sending the quotation email is a separate, non-blocking step. If it fails, the quotation and credits stay valid and the email is retried, because reversing credits the client has already been quoted for is the worse outcome.
- Crediting several companies happens in one atomic transaction, so a partial top-up cannot occur.
- Each submission carries an idempotency key, so a retry or double click cannot produce a second quotation or double credits.

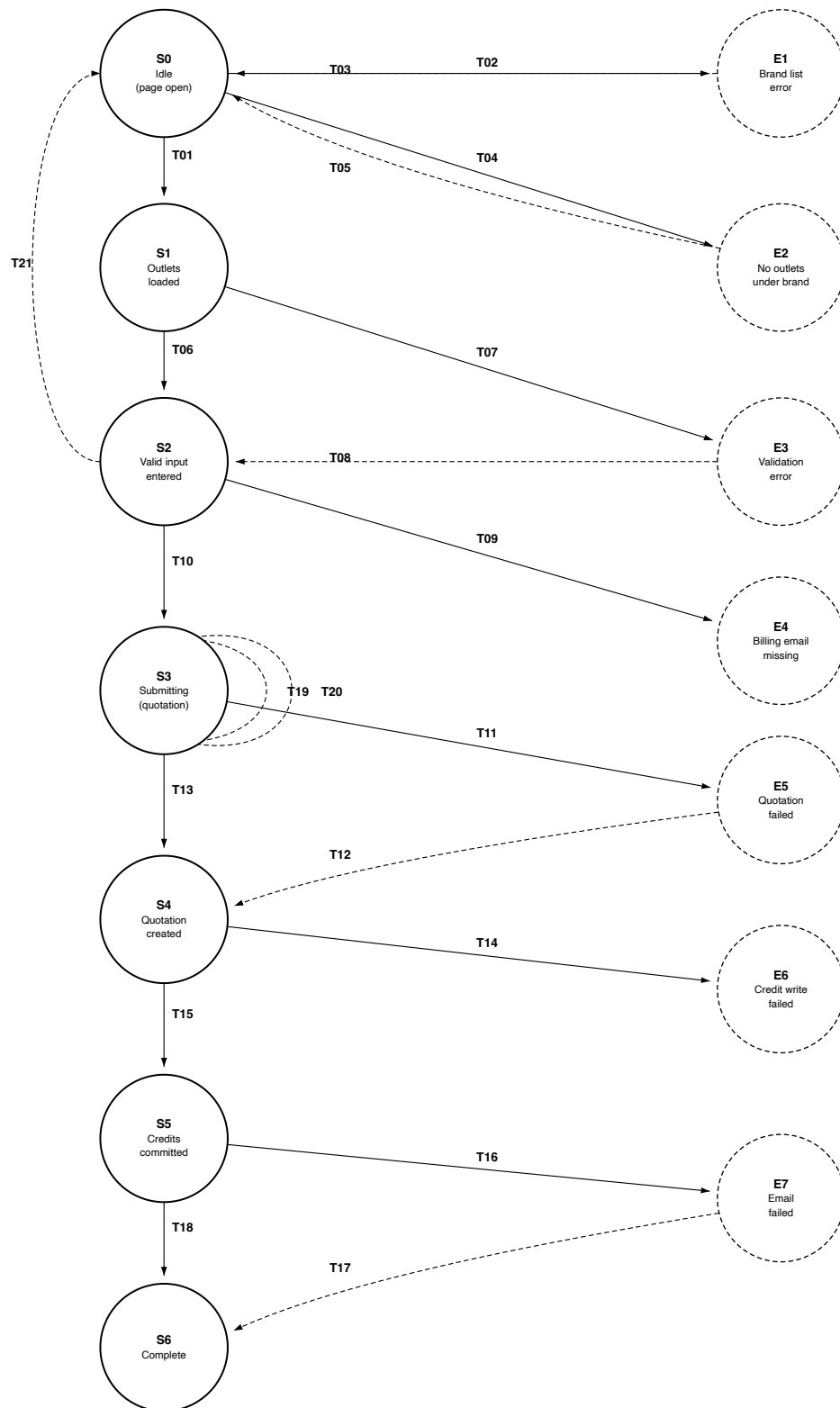
## 2. Assessment 2, Part 1 - Flowchart (including error handling and messages)



### 3. Assessment 2, Part 2 - Test cases

#### 3.1 State transition model

The feature is modelled with the four elements of state transition testing: **states** (the circles), **transitions** (the arrows), **events** (what triggers a transition) and **actions** (what the system does in response, including the message shown). Arrows are numbered; the event and action for each are listed in the transition table.



| State | Meaning                      |
|-------|------------------------------|
| S0    | Page open, no brand selected |

|       |                                                      |
|-------|------------------------------------------------------|
| S1    | Brand selected, outlets loaded                       |
| S2    | Valid top-up amounts entered                         |
| S3    | Submitting (quotation request in flight)             |
| S4    | Quotation generated                                  |
| S5    | Credits committed                                    |
| S6    | Complete (emailed, audit logged, success screen)     |
| E1-E7 | Error states, one per failure point in the flowchart |

### 3.2 Transition table

| ID  | From | Event                                         | Action (system response / message)                                         | To |
|-----|------|-----------------------------------------------|----------------------------------------------------------------------------|----|
| T01 | S0   | Select a brand that has outlets               | Show outlet list with company (billing entity) and current balance         | S1 |
| T02 | S0   | Open page, brand list fails to load           | ERR-01 "Unable to load brands. Please try again." + Retry                  | E1 |
| T03 | E1   | Retry after service restored                  | Brand list displayed                                                       | S0 |
| T04 | S0   | Select a brand with no outlets                | ERR-02 "No outlets found under this brand."                                | E2 |
| T05 | E2   | Return to brand selection                     | Selection cleared                                                          | S0 |
| T06 | S1   | Enter valid top-up amounts                    | NEXT enabled                                                               | S2 |
| T07 | S1   | Enter invalid amount                          | ERR-03 inline message; entered values kept                                 | E3 |
| T08 | E3   | Correct the input                             | Error cleared, NEXT enabled                                                | S2 |
| T09 | S2   | Click NEXT with missing/invalid billing email | ERR-04 naming the company; no quotation, no credits                        | E4 |
| T10 | S2   | Click NEXT with all data valid                | Button locked, idempotency key created                                     | S3 |
| T11 | S3   | QuickBooks error or timeout                   | ERR-05 "No credits have been added." + Retry                               | E5 |
| T12 | E5   | Retry with the same idempotency key           | Exactly one quotation created                                              | S4 |
| T13 | S3   | Quotation generated                           | Quotation number issued, line item per company                             | S4 |
| T14 | S4   | Credit transaction fails                      | ERR-06, transaction rolled back, quotation voided/flagged, finance alerted | E6 |
| T15 | S4   | Credits committed atomically                  | balance = balance + top-up, per company                                    | S5 |
| T16 | S5   | Quotation email fails to send                 | WARN-01; credits and quotation remain valid; Resend offered                | E7 |
| T17 | E7   | Resend (manual or queued)                     | Email delivered; credits not applied again                                 | S6 |
| T18 | S5   | Quotation email sent                          | Audit log written; success screen shown                                    | S6 |
| T19 | S3   | Second submit (double click)                  | Blocked by idempotency key; no duplicate quotation or credits              | S3 |
| T20 | S3   | Session expires mid-submission                | Re-authentication prompt; flow resumes once, single credit                 | S3 |
| T21 | S2   | Cancel or navigate away                       | All input discarded; no quotation, no credits, no audit entry              | S0 |

Every transition listed above is covered by exactly one test case, and no transition is tested twice. T07 is a single transition reached by six different invalid inputs, so it is written as one data-driven test case (TC-07) rather than six near-identical ones.

Transition to test case: T01 - TC-01, T02 - TC-02, T03 - TC-03, T04 - TC-04, T05 - TC-05, T06 - TC-06, T07 - TC-07, T08 - TC-08, T09 - TC-09, T10 - TC-10, T11 - TC-11, T12 - TC-12, T13 - TC-13, T14 - TC-14, T15 - TC-15, T16 - TC-16, T17 - TC-17, T18 - TC-18, T19 - TC-19, T20 - TC-20, T21 - TC-21. Business rules R1 to R6 are covered by TC-22 to TC-27.

### 3.3 Business rules (traceability)

| ID | Rule                                                                                                |
|----|-----------------------------------------------------------------------------------------------------|
| R1 | SMS credits are charged and credited to the company (billing entity), never to the brand as a whole |
| R2 | A single submission may cover outlets belonging to different companies                              |

|    |                                                                                             |
|----|---------------------------------------------------------------------------------------------|
| R3 | Only outlets with a top-up entered are quoted and credited                                  |
| R4 | Quotation line totals, tax and grand total match the system calculation to 2 decimal places |
| R5 | A top-up increments the existing balance rather than replacing it                           |
| R6 | Every successful top-up is recorded in the audit log                                        |

### 3.4 Test cases

Each case is traceable to a transition (T) or business rule (R), and states its own preconditions and data so it can be repeated by anyone.

| ID    | Ref | Title                                          | Precondition                                        | Steps                                                    | Expected result                                                                           | Type     |
|-------|-----|------------------------------------------------|-----------------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|----------|
| TC-01 | T01 | Select a brand that has outlets                | Brand 'SilverTree' has 3 outlets across 2 companies | Select 'SilverTree'                                      | Outlet list shows all 3 outlets with outlet name, company and current SMS balance         | Positive |
| TC-02 | T02 | Brand list fails to load                       | Brand list service unavailable                      | Open the Top-Up page                                     | ERR-01 shown with Retry. No data written anywhere                                         | Negative |
| TC-03 | T03 | Recover from brand list failure                | ERR-01 displayed; service restored                  | Click Retry                                              | Brand list loads; back in S0                                                              | Recovery |
| TC-04 | T04 | Brand with no outlets                          | Brand 'EmptyBrand' has 0 active outlets             | Select 'EmptyBrand'                                      | ERR-02 shown; NEXT unavailable                                                            | Negative |
| TC-05 | T05 | Return from the no-outlet state                | ERR-02 displayed                                    | Choose another brand                                     | Selection cleared and the new brand's outlets load                                        | Recovery |
| TC-06 | T06 | Enter valid top-up quantities                  | Outlets displayed                                   | Enter 500 for outlet A, 1000 for outlet B, leave C blank | Values accepted; NEXT enabled                                                             | Positive |
| TC-07 | T07 | Invalid input (data-driven, single transition) | Outlets displayed                                   | Enter in turn: 0 / -50 / 1.5 / abc / all blank / 100001  | Matching ERR-03 message each time; NEXT stays disabled; valid values preserved            | Negative |
| TC-08 | T08 | Correct an invalid input                       | ERR-03 displayed                                    | Replace the invalid value with 500                       | Error clears; NEXT enabled                                                                | Recovery |
| TC-09 | T09 | Missing billing email                          | Outlet B's company has no billing email             | Enter valid amounts, click NEXT                          | ERR-04 naming the company; no quotation, no credits                                       | Negative |
| TC-10 | T10 | Submit a valid top-up                          | Valid amounts; all billing emails present           | Click NEXT                                               | Button locks; idempotency key created; quotation requested                                | Positive |
| TC-11 | T11 | QuickBooks unavailable                         | QuickBooks returns error/timeout                    | Click NEXT                                               | ERR-05 shown; no quotation; no credits on any company                                     | Negative |
| TC-12 | T12 | Retry after QuickBooks failure                 | ERR-05 displayed; QuickBooks restored               | Click Retry                                              | Exactly one quotation exists; credits applied once                                        | Recovery |
| TC-13 | T13 | Quotation generated                            | Valid submission; QuickBooks healthy                | Click NEXT                                               | Quotation created with one line item per company matching the entered amounts             | Positive |
| TC-14 | T14 | Credit write fails after quotation             | Quotation created; credit transaction fails mid-way | Complete submission                                      | ERR-06; no company partially credited; quotation voided/flagged; finance alerted          | Negative |
| TC-15 | T15 | Credits committed per company                  | Quotation created successfully                      | Complete submission                                      | Each company's balance rises by its own amount; other companies unchanged                 | Positive |
| TC-16 | T16 | Quotation email fails                          | Credits committed; mail service down                | Complete submission                                      | WARN-01; credits and quotation remain valid; Resend offered                               | Negative |
| TC-17 | T17 | Resend the quotation email                     | WARN-01 displayed; mail service restored            | Click Resend                                             | Email delivered; credits not applied a second time                                        | Recovery |
| TC-18 | T18 | Full happy path                                | All services healthy                                | Complete the flow                                        | Email sent, audit log written, success screen shows quotation number and updated balances | Positive |
| TC-19 | T19 | Double submission                              | Valid amounts entered                               | Click NEXT twice rapidly                                 | One quotation and one credit transaction only                                             | Negative |

|       |     |                                         |                                          |                                                  |                                                                                       |               |
|-------|-----|-----------------------------------------|------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------|---------------|
| TC-20 | T20 | Session expires during submission       | Submission in flight; token expires      | Complete submission                              | Re-authentication prompt; afterwards exactly one quotation and one set of credits     | Negative      |
| TC-21 | T21 | Cancel mid-flow                         | Amounts entered, NEXT not clicked        | Navigate away, then return                       | No quotation, no credits, no audit entry; clean form                                  | Negative      |
| TC-22 | R1  | Credits are company-scoped              | Brand spans Company X and Company Y      | Top up an outlet belonging to Company X only     | Only Company X is credited and billed; Company Y untouched                            | Business rule |
| TC-23 | R2  | Several companies in one submission     | Brand spans Company X and Company Y      | Top up 500 for X's outlet and 800 for Y's outlet | Charges separated per billing entity, each emailed to its own address; X +500, Y +800 | Business rule |
| TC-24 | R3  | Partial fill of the outlet list         | Brand has 3 outlets                      | Fill outlet A only                               | Only A's company is quoted and credited; B and C excluded entirely                    | Business rule |
| TC-25 | R4  | Quotation totals and rounding           | Amounts producing a fractional tax value | Submit                                           | Line totals, tax and grand total match to 2 decimal places with no rounding drift     | Business rule |
| TC-26 | R5  | Balance is incremented, not overwritten | Company X holds 2,000 credits            | Top up 500                                       | Balance becomes 2,500                                                                 | Business rule |
| TC-27 | R6  | Audit trail completeness                | Successful top-up                        | Complete flow, open audit log                    | Entry records user, brand, companies, amounts, quotation number and timestamp         | Business rule |

#### 4. Highest-risk areas

- **Money and credits out of step (TC-11, TC-12, TC-14, TC-19, TC-20).** The expensive failure is a client billed without credits, or credited twice. The idempotency key and the atomic transaction are what prevent it.
- **Crediting the wrong entity (TC-22, TC-23, TC-24).** The brand-to-company split is the subtlest rule in the feature, so it gets dedicated tests.
- **Silent partial success (TC-14, TC-16).** The user must always be told exactly which of quotation, credits and email succeeded.